ANALYSIS REPORT

Web Scraping, Cleaning and Temporal Analysis of Allbirds Product Data

DSC2107 – Data Mining and Wrangling — Group 9 — 15–19 October 2026

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1. Data Summary

The analysis uses the cleaned SQLite database `data/allbirds.db`, containing **692 unique products** and **52,178 variant observations** across five distinct collection dates (15, 16, 17, 18 and 19 October 2026).

There are **zero missing prices** and **zero missing identifiers**. The price range is **0.80–165** (mean 101.84, median 105, interquartile range 75–130).

A total of **35,772** variant observations were unavailable and **16,406** were available, giving an overall availability rate of about **31.5%**.

2. Price Distribution and Outliers (Qa)

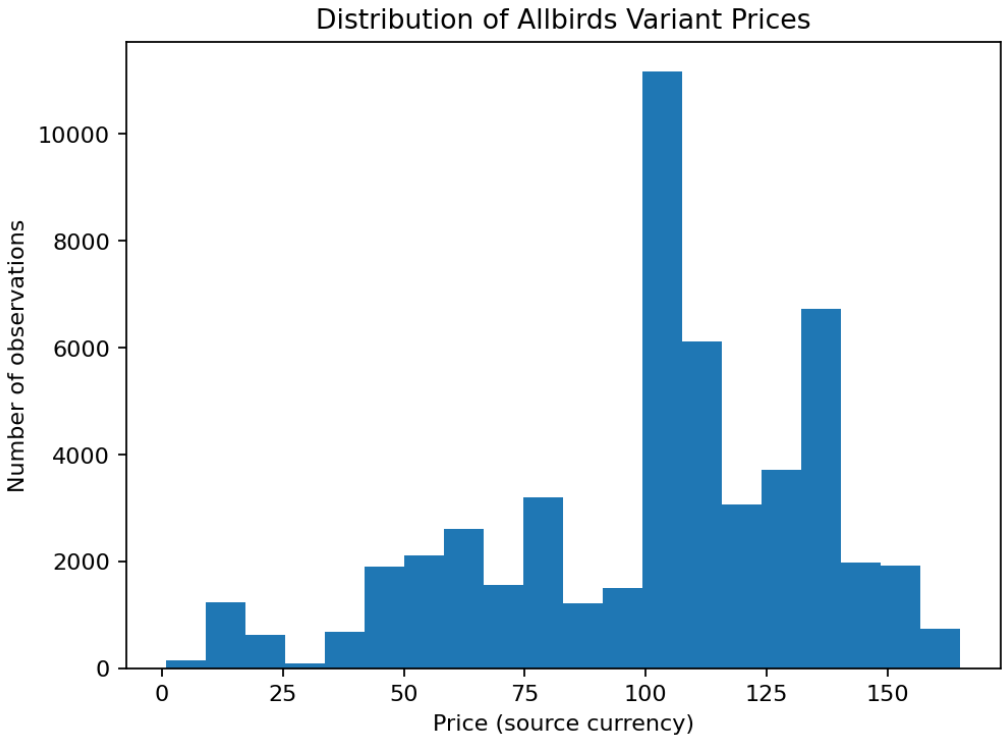


Figure 1: Distribution of Allbirds variant prices. Median 105, with a long left tail to 0.80 and a right tail to 165.

Prices are **multimodal**: the largest cluster sits around 100–110, with secondary clusters near 75 and 130–140.

The two **outliers** are:

- **Minimum 0.80** – possibly a placeholder, rounding artefact, or add-on product. This value should be investigated before category-level analysis.
- **Maximum 165** – a premium price point that may be legitimate.

The low minimum may pull the mean (101.84) below the median (105).

3. Average Price by Product Type (Qa Continued)

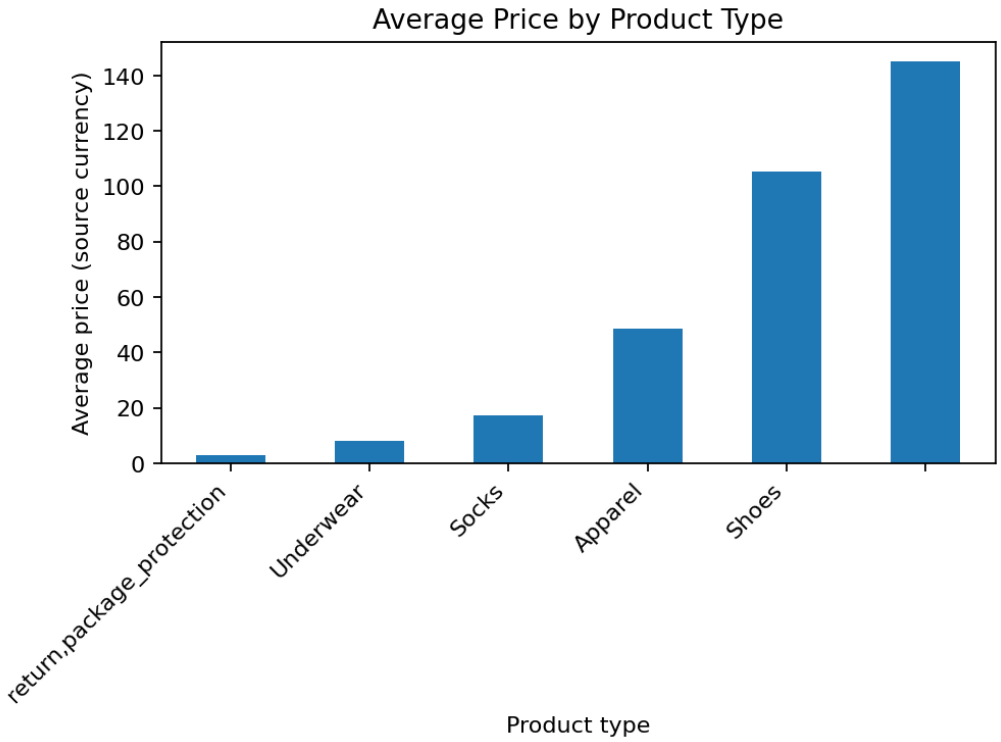


Figure 2: Average variant price by product type. The blank category and the return/package-protection entry should be investigated before category comparisons.

Shoes dominate the product mix (602 of 692 products). Socks and Underwear are cheaper than shoes in the recorded data.

The blank product type and `return,package_protection` category should be treated as data-quality issues and excluded from normal product-type comparisons where appropriate.

4. Availability Overall and Over Time (Qb, Qc)

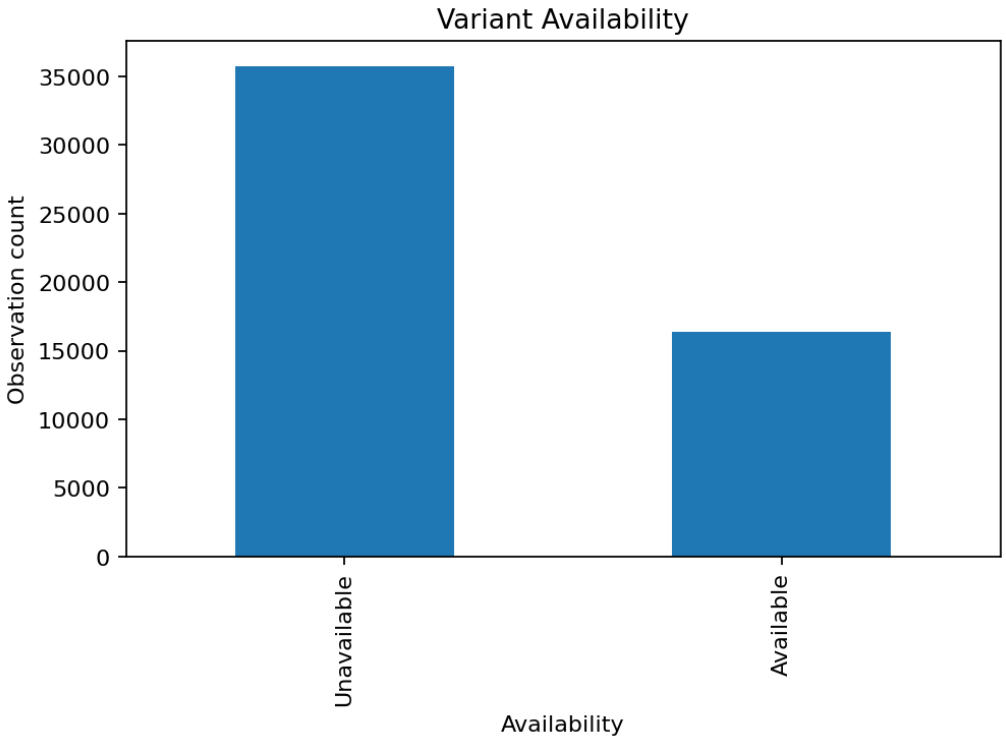


Figure 3: Variant availability counts across all observations. Approximately 31.5% of observations were available.

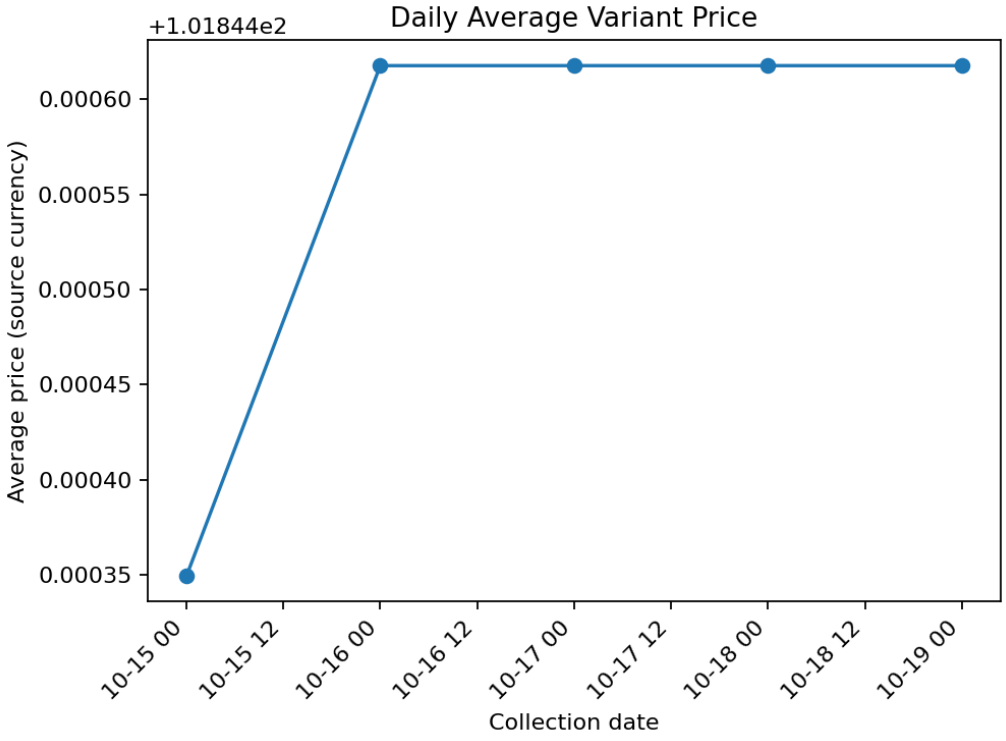


Figure 4: Daily average variant price across the five collection dates.

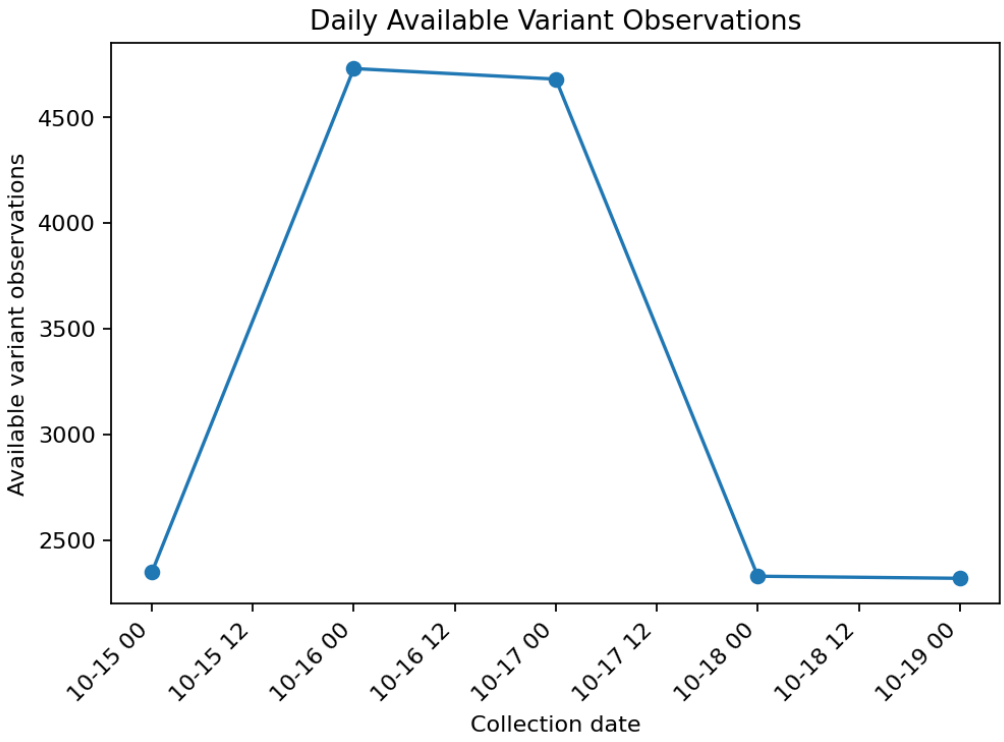


Figure 5: Daily available variant observations. Counts are affected by the number of scrapes conducted on each date.

The availability count is higher on 16–17 October because two scrapes were run on each of those dates. The percentage available is more suitable for comparing dates with different numbers of observations.

The recorded availability percentage remained approximately 31.1–31.7% across the five dates. Average price was reported to be broadly stable at around 101.84.

5. Price Versus Availability (Qb Adapted)



Figure 6: Price distribution for available versus unavailable variants.

The median price is reported to be similar for available and unavailable variants, at around 105. The available group has a longer lower-price tail, extending to approximately 1–10.

This may reflect cheaper accessory products, but the chart alone does not establish that accessories are more likely to remain in stock or that premium shoes are more likely to sell out. Further analysis by product type would be needed to test that relationship.

6. Questions Not Answerable from This Feed

The assignment asks about **ratings**, **review counts**, **rankings**, and **discounts**.

The Allbirds `products.json` feed used in this project does not expose ratings, review counts, or rankings, so those questions cannot be fully answered using this feed alone.

Discounts may be investigated using `compare_at_price`, but that field was not extracted in the current pipeline. It should be added in a follow-up iteration to support discount analysis.

7. Temporal Change (Our Own Question)

Comparing the first and last observation per variant showed that **70 of 7,454 variants** (about 0.94%) changed availability status during the five-day period, while 7,384 did not.

This is a small short-term change signal. The result should be interpreted in the context of the limited collection period and the available observations.

8. Conclusions

1. The recorded price distribution is concentrated between 75 and 130, with a minimum value of 0.80 that should be investigated.
2. Availability was approximately 31.5% across the five collection dates.
3. The price and availability distributions can be explored further by product type to test whether price is associated with availability.
4. 70 of 7,454 variants changed availability status between their first and last valid observations.
5. Ratings, review counts, rankings, and discounts cannot be fully analysed from the current feed and pipeline.